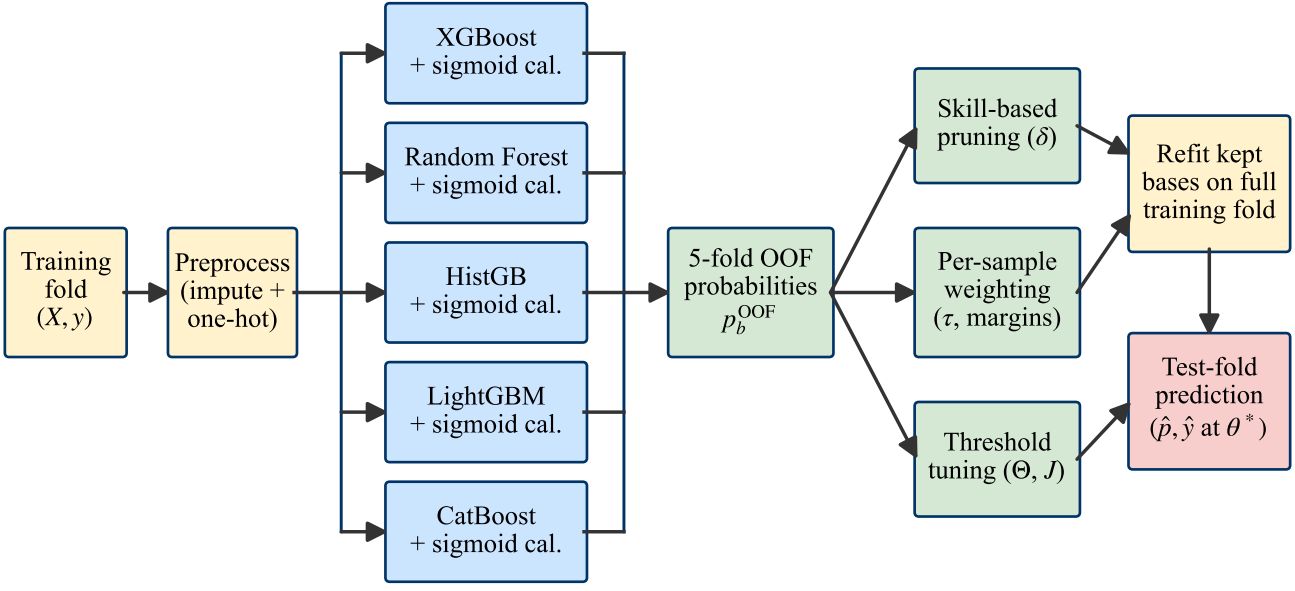
Training fold (X, y)
Preprocess (impute + one-hot)
XGBoost + sigmoid cal.
Random Forest + sigmoid cal.
HistGB + sigmoid cal.
LightGBM + sigmoid cal.
CatBoost + sigmoid cal.
5-fold OOF probabilities p_b^{OOF}
Skill-based pruning (δ)
Per-sample weighting (τ, margins)
Threshold tuning (Θ, J)
Refit kept bases on full training fold
Test-fold prediction ($\hat{p}, \hat{y}$ at θ^*)